

# The Body Basher Quick Start Guide

## What is the Body Basher?

The Body Basher is a high voltage precision square pulse generator. It's primary use is for performing structured forward and reverse body bias attacks on depcapped targets. It is capable of generating positive and negative pulses of up to 500V with widths from 20ns upwards. A an off-the-shelf 9V-12V, 35W DC PSU is needed. The device is controlled through a USB interface and several digital inputs and outputs.

## Setting it up

The setup of the Body Basher should be **mounted with 3xM2.5** screws using the mounting holes around the output capacitor. An additional mounting hole, located near the digital I/O pins may be used for extra mechanical stability. A **9V-12V, 35W DC power supply** is connected to the green Phoenix connector. A data micro-USB cable should be used to connect the Body Basher to the control computer. The two pins labelled PULSE should be connected to a device that will generate a low voltage control pulse. An optional signal V\_READY may be used to detect when the Body Basher is in an armed state. **Take note about the irregular arrangement of GND pins - see the figure!** On the target side - the pins labelled COM should be connected to the target ground.

## Using the Body Basher

The Body Basher is controlled through a USB virtual COM port at a recommended 115200 baud, XON/XOFF flow control should be on. The list of all commands with detailed instructions can be accessed using the *help* command issued through the terminal. For convenience a list of commands is presented here as well.

| CMD    | Options      | Description                                                  |
|--------|--------------|--------------------------------------------------------------|
| help   | /            | list of functions with instructions                          |
| arm    | -v -p (-t)   | charges the output capacitor according to the given settings |
| disarm | /            | discharges the capacitor                                     |
| status | /            | prints current device information                            |
| error  | -p -c -h     | error status and control                                     |
| tps    | -rr -wr -val | TPS55289 register write and read                             |

After issuing the comand 1:

$$\text{arm -v <voltage> -p <0/1> >(-t <tolerance>)} \quad (1)$$

It will take a maximum of 500ms for the Body Basher to prepare the output. At that time, the V\_READY pin is set high, an *Armed!* notification is issued in the terminal and LED **D1** turns on. From that point on, the PULSE input pin is used to generate a pulse on the output, corresponding to the previously determined settings. The device can be disarmed using the *disarm* command.

The Body Basher uses a 10μF output capacitor to provide energy for the pulse. This capacitor is not recharged unless an arm command is reissued. This means that when injecting many or longer pulses, the device needs to be rearmed periodically. The device supports consecutive arm commands of any voltage or polarity.

## Specifications

| Specification          | Value                       |
|------------------------|-----------------------------|
| Maximum output voltage | ±500V                       |
| Minimum output voltage | ±50V                        |
| Maximum output current | ±3.5A                       |
| Minimum pulse width    | 20ns                        |
| Maximum pulse width    | None: exponential discharge |
| PULSE-to-output delay  | ≈ 55ns                      |
| Rise time              | ≈ 3.5ns                     |
| Fall time              | ≈ 3.5ns                     |

## Safety considerations

**Electrical:** This device operates at high voltages capable of causing serious injury or death.

**UNDER NO CIRCUMSTANCE is any exposed metal to be touched while the device is armed** - especially under the isolation line on the diagram. Disarm or power off to make setup adjustments.

**Thermal:** The bleeder resistor may get warm to the touch if the device is held in the armed state for extended periods.

## Troubleshooting

Faults may occur for various reasons such as accidents, improper use or component failure. The Body Basher is designed to detect and handle some that may pose a danger to people, property or itself. These include:

| Cause                          | Error code |
|--------------------------------|------------|
| Arming timeout                 | 0x01       |
| Arming over-voltage            | 0x02       |
| LMG2610 over-temperature fault | 0x04       |
| Output over-current            | NA         |
| Short-circuit protection       | NA         |

The current error code may be checked using command *error*. The error code that is printed is bitwise *OR* of all the detected errors since power-up. It does not mean that those error states are necessarily still valid. Regardless, if the device detects an error, it will enter the *ERROR* state, discharge the capacitor and light **D1 and D2** LEDs. The errors may be cleared using the *error -c* command.

Errors without an error code are not logged by the controller as they are features of the components that are used in the device.

If the Body Basher firmware finds itself in an undefined state, it will automatically power cycle. This may occur if its output buffer gets full.

## Contact Information

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## Body Basher Diagram

